ENDG 319

Probability, Statistics and Machine Learning

Lecture 9: Bernoulli and Binomial Distributions

Commonly used distributions • Counts • Sample proportions

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Learning outcomes for Lecture 9

ONE BINARY TRIAL → A COUNT OF SUCCESSES → AN ESTIMATE OF p

1	**Code**	Represent a two-outcome trial with a Bernoulli random variable.
2	**Count**	Recognize a binomial experiment and construct its PMF.
3	**Calculate**	Find probabilities, cumulative probabilities, means, and variances.
4	**Estimate**	Use a sample proportion to estimate an unknown success probability.

> Define “success” and check the trial assumptions before choosing a distribution.

A probability distribution describes a population

CHOOSE THE VARIABLE THAT REPRESENTS ONE OBSERVATION

DISCRETE

Defective or nondefective items

Assign a number to each category.
Describe the probabilities with a PMF.

CONTINUOUS

Lengths of fish in a lake

Represent each length by a value of X.
Describe the distribution with a PDF.

Standard families provide models for recurring kinds of random variables.

Binary coding determines the success probability

OPENING EXAMPLE: 10% OF THE ITEMS ARE DEFECTIVE

Select one item at random. Use the coding specified in the opening example:

Item status	Value of X	Probability
Defective	0	0.10
Nondefective	1	0.90

Keep the event coded as 1 in view: it will define p for a Bernoulli model.

> Here X = 1 means nondefective, so P(X = 1) = 0.90.

The Bernoulli distribution: one two-outcome trial

A “SUCCESS” IS THE OUTCOME CODED AS 1

$$X \sim \text{Bernoulli}(p)$$

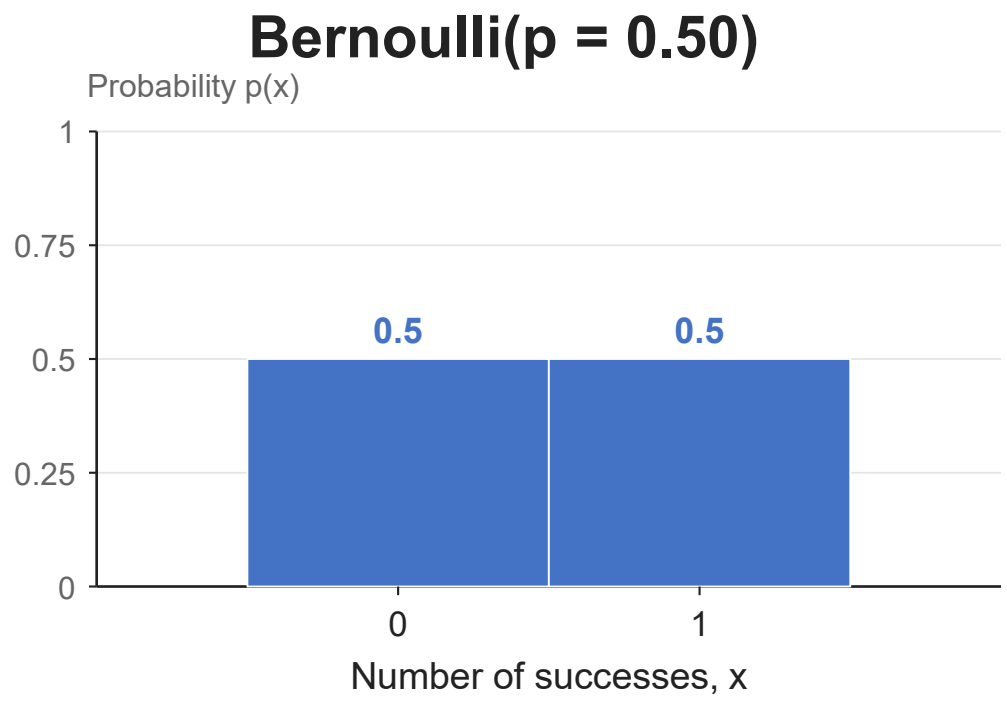
Outcome	Value x	p(x) = P(X = x)
Failure (F)	0	1 − p
Success (S)	1	p

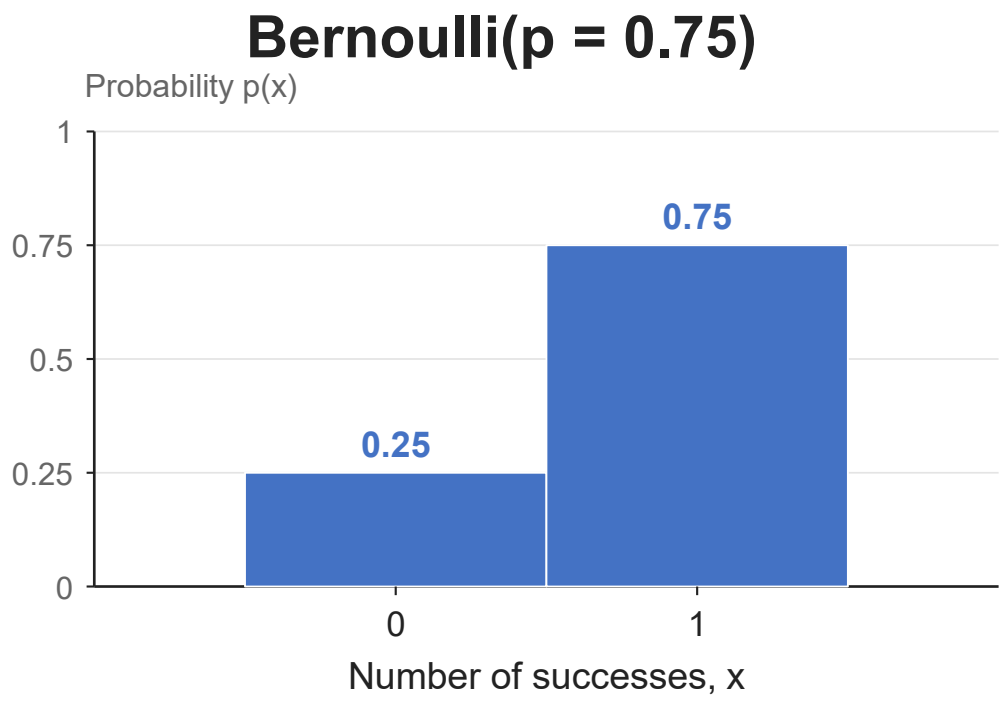
Parameter: $0 \le p \le 1$. Only the values 0 and 1 are possible.

> “Success” is a mathematical label; it need not be a desirable engineering outcome.

Bernoulli probability histograms

THE SAME TWO POSSIBLE VALUES; DIFFERENT PROBABILITY WEIGHTS





Unit-width bars: each rectangle's area equals its probability mass.

Increasing p shifts probability mass from x = 0 to x = 1.

Bernoulli mean and variance

PROBABILITY-WEIGHT THE VALUES 0 AND 1

MEAN

$$E(X) = 0(1-p) + 1(p) = p$$

SECOND MOMENT

$$E(X^2) = 0^2(1-p) + 1^2 p = p$$

VARIANCE

$$\mathrm{Var}(X) = p - p^2 = p(1-p)$$

For a Bernoulli variable, $\mu = p$ and $\sigma^2 = p(1-p)$.

Example 1: a defective component

EXAMPLE 1 NOW CODES A DEFECT AS X = 1

Ten percent of the components from a manufacturing process are defective. Choose one component at random.

X = 1 if the component is defective; X = 0 otherwise.

What is the distribution of X? Find its mean and variance.

p is the probability of the event coded as X = 1.

Example 1: match p to the event being counted

DEFECTIVE IS THE CODED SUCCESS IN THIS EXAMPLE

x	Interpretation	p(x)
0	Nondefective	0.90
1	Defective	0.10

$$X \sim \text{Bernoulli}(0.10)$$

$$\mu_X = p = 0.10$$

$$\sigma_X^2 = (0.10)(0.90) = 0.09$$

Changing the 0/1 coding changes which probability is called p.

The binomial distribution: check four conditions

X COUNTS SUCCESSES IN n BERNOULLI TRIALS

1	**Fixed number**	The number of trials n is specified in advance.
2	**Two coded outcomes**	Each trial is classified as success or failure.
3	**Independent trials**	Knowing other trial outcomes does not change a trial's probability.
4	**Same success probability**	Every trial uses the same value of p.

A count is binomial only when the trial model supports all four conditions.

One trial versus a count over many trials

SAME SUCCESS DEFINITION; A DIFFERENT RANDOM VARIABLE

For trial i, define $I_i = 1$ on success and $I_i = 0$ on failure.

$$I_i \sim \text{Bernoulli}(p)$$

Independent trials with the same p

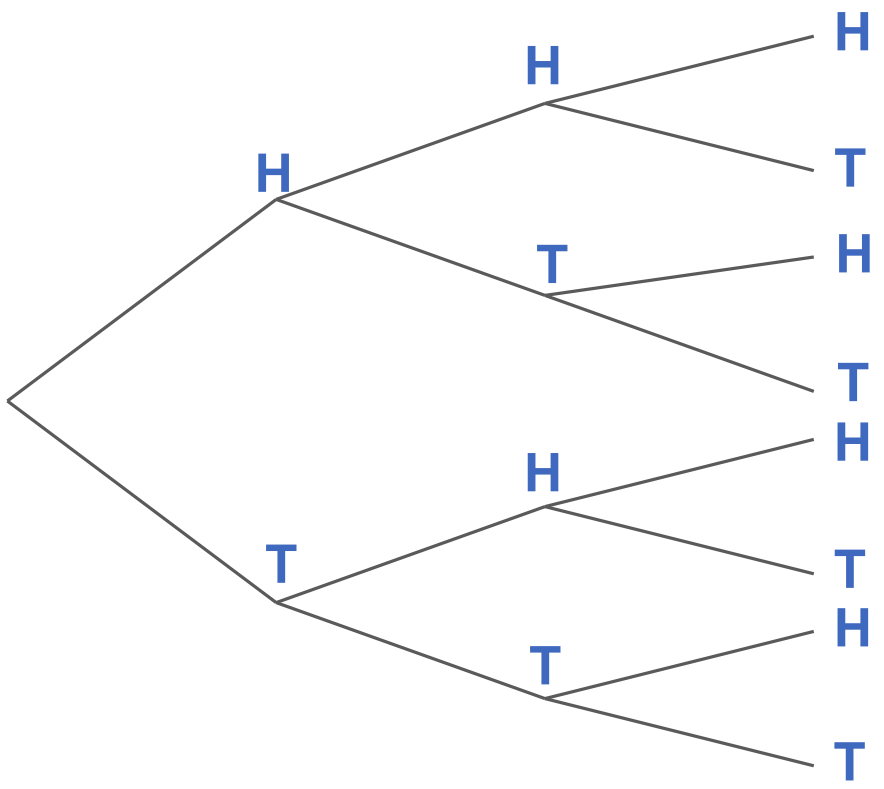
$$X = I_1 + I_2 + \cdots + I_n \sim \text{Bin}(n, p)$$

A Bernoulli variable is 0 or 1. A binomial count can be any integer from 0 to n.

Three fair-coin tosses: list the complete outcomes

n = 3, p = 0.50; X IS THE NUMBER OF HEADS

Independent tosses
H = success
T = failure



Outcome	x
HHH	3
HHT	2
HTH	2
HTT	1
THH	2
THT	1
TTH	1
TTT	0

A complete outcome is a three-toss sequence; X records only its number of heads.

A single sequence is not the whole event X = 1

EXACTLY ONE HEAD CAN OCCUR IN THREE DIFFERENT POSITIONS

H T T	T H T	T T H

$$P(HTT) = p(1-p)^2 \qquad P(X=1) = 3p(1-p)^2$$

$$p = \tfrac{1}{2}: \qquad P(X=1) = 3\left(\tfrac{1}{2}\right)^3 = \tfrac{3}{8}$$

Multiply along one sequence; add over all sequences that give the requested count.

Count the positions occupied by successes

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CHOOSE x OF THE n TRIAL POSITIONS

The number of sequences containing x successes and n − x failures is

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Equivalent view: arrange x identical S labels and n − x identical F labels.

$$n = 3,\ x = 1: \qquad \binom{3}{1} = \frac{3!}{1!\,2!} = 3$$

Success positions determine the sequence, so use combinations—not permutations.

The binomial PMF combines counting and probability

X ~ Bin(n, p)

$$p(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

NUMBER OF SEQUENCES

$$\binom{n}{x}$$

PROBABILITY OF ONE SEQUENCE

$$p^x (1-p)^{n-x}$$

x = 0, 1, …, n; p(x) = 0 otherwise.

Count the qualifying sequences, then multiply by the probability of each sequence.

Three fair tosses: construct the complete PMF

X ~ Bin(3, 0.50)

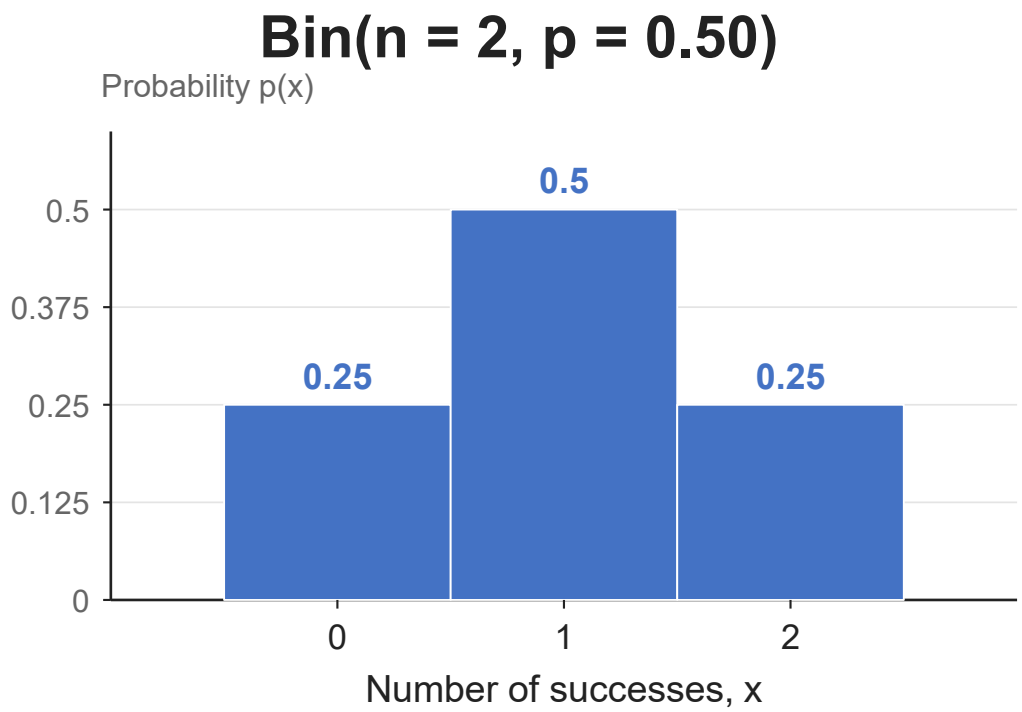
x	Sequences giving X = x	Number	p(x)
0	TTT	1	1/8 = 0.125
1	HTT, THT, TTH	3	3/8 = 0.375
2	HHT, HTH, THH	3	3/8 = 0.375
3	HHH	1	1/8 = 0.125

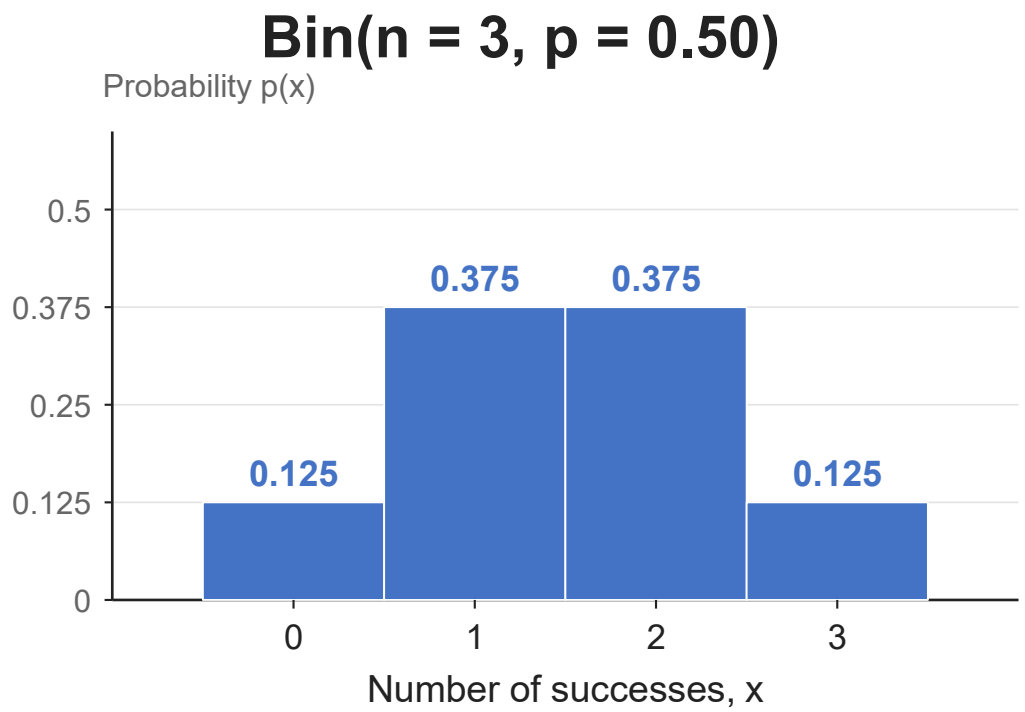
$$\sum_{x=0}^{3} p(x) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$$

The four possible counts are not equally likely, even though the eight sequences are.

Binomial probability histograms

KEEP p = 0.50; CHANGE THE NUMBER OF TRIALS





Both distributions are symmetric; their probabilities each sum to 1.

For n = 3, the middle probabilities are 3/8 = 0.375—not 0.50.

Binomial mean, variance, and standard deviation

X ~ Bin(n, p): ADD n INDEPENDENT BERNOULLI INDICATORS

MEAN

$$\mu_X = np$$

VARIANCE

$$\sigma_X^2 = np(1-p)$$

STANDARD DEVIATION

$$\sigma_X = \sqrt{np(1-p)}$$

For three fair tosses: $\mu = 1.5$ $\sigma^2 = 0.75$ $\sigma \approx 0.8660$

A mean describes an average count; it need not be a possible integer outcome.

Bernoulli and binomial: related, but not interchangeable

THE SUCCESS PROBABILITY p HAS THE SAME MEANING IN BOTH MODELS

Feature	Bernoulli(p)	Bin(n, p)
What X records	One coded trial result	Number of successes in n trials
Possible values	0, 1	0, 1, …, n
Mean	p	np
Variance	p(1 − p)	np(1 − p)

Bernoulli(p) is the binomial model with n = 1.

PMF versus cumulative binomial distribution

EXACTLY x AND AT MOST x ARE DIFFERENT EVENTS

PMF: “exactly x”

$$p(x) = P(X = x)$$

CDF: “at most x”

$$F(x) = P(X \leq x)$$

For an integer k with 0 ≤ k ≤ n:

$$F(k) = \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j}$$

The CDF adds all probability masses at or below its threshold.

Translate probability wording before calculating

LET k BE AN INTEGER; $F(k) = P(X \le k)$

Words	Event	Calculation
Exactly k successes	$X = k$	$p(k)$
At most / no more than k	$X \le k$	$F(k)$
Fewer than k	$X < k$	$F(k - 1)$
At least k	$X \ge k$	$1 - F(k - 1)$
More than k	$X > k$	$1 - F(k)$

For an integer count, changing < to ≤ may add a whole probability mass.

Example 2: no more than two sixes in eight rolls

A FAIR DIE IS ROLLED EIGHT TIMES

Find the probability that no more than two sixes occur.

Success: roll a six p = 1/6
Failure: roll anything else 1 − p = 5/6

Assume independent rolls. Let X count the sixes.

$$X \sim \text{Bin}(8,\, 1/6), \qquad \text{find } P(X \leq 2)$$

The two coded outcomes are “six” and “not six”—not the six individual face values.

Example 2: add the three disjoint possibilities

P(X ≤ 2) = P(X = 0) + P(X = 1) + P(X = 2)

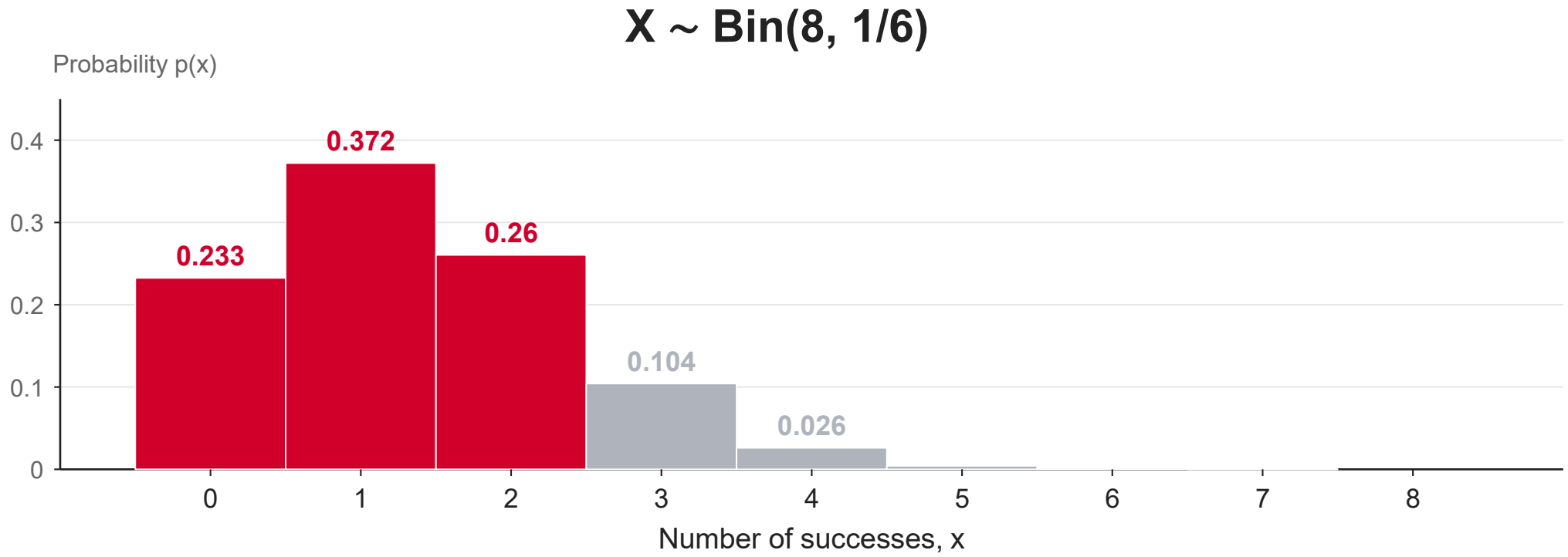
x	Binomial PMF term	Probability
0	C(8,0)(1/6)⁰(5/6)⁸	0.232568…
1	C(8,1)(1/6)¹(5/6)⁷	0.372109…
2	C(8,2)(1/6)²(5/6)⁶	0.260476…

$$P(X \le 2) = \sum_{j=0}^{2} \binom{8}{j} \left(\frac{1}{6}\right)^{j} \left(\frac{5}{6}\right)^{8-j}$$

P(X ≤ 2) ≈ 0.8652: about 86.52% under the independent fair-die model.

Example 2: visualize the cumulative probability

SHADE THE COUNTS x = 0, 1, AND 2



Red bars: P(X ≤ 2) ≈ 0.8652

Other counts: P(X > 2) ≈ 0.1348

"No more than two" includes zero, one, and two successes.

Use a cumulative binomial table carefully

A DIFFERENT EXAMPLE: X ~ Bin(8, 0.50); FIND P(X < 3)

1 Locate n = 8. 2 Use the row k = 2. 3 Read the p = 0.50 column.

n = 8: k	p = 0.25	p = 0.50	p = 0.75
0	0.100113	0.003906	0.000015
1	0.367081	0.035156	0.000381
2	**0.678543**	**0.144531**	**0.004227**
3	0.886185	0.363281	0.027298

Computed classroom excerpt: entries are F(k) = P(X ≤ k), not individual PMF values.

F(2) = 37/256 = 0.14453125 ≈ 0.145

X < 3 means X ≤ 2, so read F(2)—not F(3).

When p is unknown, estimate it from a sample

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CONDUCT n INDEPENDENT TRIALS WITH A COMMON UNKNOWN p

Count the successes, X, and divide by the number of trials, n.

$$\hat{p} = \frac{\text{number of successes}}{\text{number of trials}} = \frac{X}{n}$$

p: unknown success probability

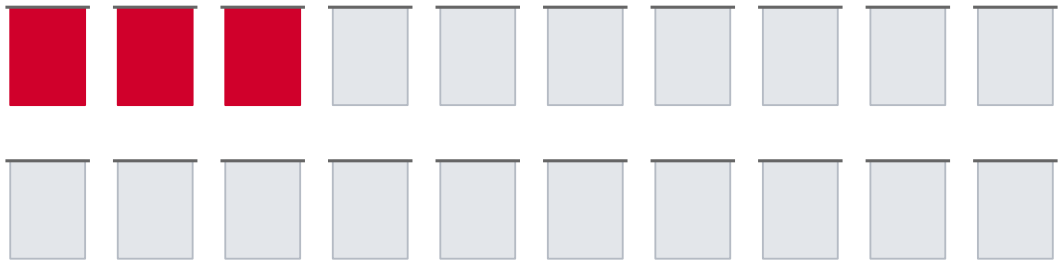
p̂: sample-based estimate

p is a population/model parameter; p̂ is computed from the observed sample.

Application: underfilled ice-cream containers

20 CONTAINERS INSPECTED; 3 ARE UNDERFILLED

Define a success as finding an underfilled container.



3 underfilled 17 not underfilled

$$\hat{p} = \frac{3}{20} = 0.15$$

Sample result: 15%

The estimated underfill probability is 0.15; the true process probability is still unknown.

Concept check: choose the event and the model

USE ONLY THE EXAMPLES ALREADY DISCUSSED

A 10% defective; X = 1 means nondefective.
What is the Bernoulli parameter p?

B Three independent fair tosses.
Why is P(X = 1) not just $(0.5)^3$?

C X ~ Bin(8, 0.5).
Express P(X ≥ 3) using the CDF.

D 3 of 20 containers are underfilled.
Does $\hat{p} = 0.15$ establish that p = 0.15?

Explain the definition or event behind your answer—not just a formula.

Concept check: solutions and reasoning

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RETURN TO THE SUCCESS CODE, THE COUNT EVENT, AND THE PARAMETER

A **$p = 0.90$**

X = 1 denotes a nondefective item.

B **$P(X = 1) = 3/8 = 0.375$**

HTT, THT, and TTH each have probability 1/8.

C **$P(X \geq 3) = 1 - F(2) \approx 0.8555$**

The excluded counts are 0, 1, and 2.

D **No: $\hat{p}$ is an estimate of p.**

A sample fraction need not equal the population probability.

Most errors begin with the model or event definition—not the arithmetic.

Lecture 9: key takeaways

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ONE TRIAL, A SUCCESS COUNT, AND AN ESTIMATED PROBABILITY

Bernoulli	One coded trial: $X = 0$ or 1; $\mu = p$; $\sigma^2 = p(1 - p)$.
Binomial	Count successes in n independent trials with the same p.
Probability calculation	Use the PMF for an exact count; use the CDF or a complement for a range.
Sample proportion	Estimate an unknown probability with $\hat{p} = X/n$; do not confuse $\hat{p}$ with p.

Define X → check n and p → verify assumptions → identify the event → calculate.